API versions: 1 2 3 4 5

Commands

/documentation
/login
/logout
/search/logspace/list_logspaces
/search/logspace/filter
/search/logspace/number_of_messages
/search/logspace/generate_statistics
/search/logspace/number_of_statistics_entries
/search/logspace/sum_of_statistics_values
/search/logspace/list_dynamic_columns
/version
/search/logspace/number_of_messages_by_steps
/search/logspace/list_logspaces_details
/search/logspace/validate

General request format

All requests are HTTP GETs or POSTs using the following format:

```
https://192.168.10.48/api/<API version>/<command>/<positional parameter>?<named parameter 1>=<value 1>&<named parameter 2>=<value 2>
```

Requests are served on the same interfaces on the same ports using the same SSL certificate as the web UI. Make sure that your client is able to verify and accepts the necessary SSL chain.

Note that each parameter must be URL encoded.

General response format

The response format is the following:

```
{
    'result': <command specific result>,
    'error':
    {
        'code': <error type specific error code>,
        'message': <optional error message>
    },
    'warnings':
    [
        <first warning message>,
        <second warning message>,
        ...
    ]
}
```

Response is JSON-encoded object which always contains three parts: `result` , `error` and `warnings` . If the command returns successfully `error.code` and `error.message` are null. If there are no warnings the `warnings` object is an empty array. The command references will only contain the `result` part.

Timestamps

Format: UNIX timestamp; seconds since 1 January 1970 UTC. Pay attention to timezones and daylight saving changes when converting local time to UNIX timestamp.
Precision of searching and creating statistics is 1 minute. Even if you specify more precise timestamps values will be rounded to the previous minute when specifying the beginning of an interval and the next minute when specifying the end of an interval.

Authentication

Authentication token can be acquired using the *login* command. After that, every HTTP request header must contain a *Cookie: AUTHENTICATION_TOKEN=<authentication token>* field.

Commands

Commands can be passed to SSB using the HTTP GET method over HTTPS protocol, with the exception of the login command which uses the POST method for security reasons. All positional parameters are always mandatory, named parameters are required if indicated (with asterisk (*)).

/documentation

Method: **GET**
Retrieve actual documentation of REST API.

Command specific response:

This documentation

Example using wget command:

```
wget -q -O - 'https://192.168.10.48/api/4/documentation'
```

/login

Method: **POST**
Post parameters:

username: Valid username to login into SSB (required, regular expression: "[A-Za-z0-9_.-]+")
password: Password of the user (required, regular expression: "[] 0-9a-zA-Z!"#$%&'()*+,; <=>@[\^`{|}_.\/:?\-~]+")

To gain access to SSB, the client program must be authenticated using the `login` command, where username and password is the same as used on web UI. This command will return with AUTHENTICATION_TOKEN in success.

Command specific response:

```
'0123456789abcdef'
```

Example using wget command:

```
wget -q -O - --post-data='username=<username>&password=<password>'
'https://192.168.10.48/api/4/login'
```

/logout

Method: **GET**

To terminate the client program's session, use the 'logout' command.
Note: if the timeout configured at "Basic Settings -> Management -> Session timeout" is reached and there is no activity, the session will be terminated automatically.

Command specific response:

Successful logout: `true`

Logout returns with exception (for example, the script was not logged in): `null`

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/logout'
```

/search/logspace/list_logspaces

Method: **GET**

Retrieves the list of searchable (indexed) logspaces. Note that:

> The list includes only those logspaces that the authenticated user can access.
> The list includes only searchable (indexed) logspaces.
> The list is in alphabetical order.

Command specific response:

```
['first_logspace_name', 'second_logspace_name', ...]
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/list_logspaces'
```

/search/logspace/filter

Method: **GET**
Positional parameters:

> **logspace_name**: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")
offset: Retrieve a subset of the result. The offset is the starting point of the subset. The value 0 means the first element in the result set (= no offsetting) (optional, default: "0", regular expression: "\d{1,19}")
limit: The maximum number of elements in the result set. If the whole result set has less elements than this value, then only those entries will be retrieved. More than 1000 records can be retrieved by executing several requests with consecutively increasing offset values (after an offset of 0, the next offset should be 1000, then 2000, and so on). (optional, default: "100", regular expression: "[0-9]{1,3}|1000")

The **filter** command is the main command of the Search API. It retrieves log message records from SSB. The received records can be filtered by time interval and search expression.
A partial result set can be queried by using the offset and limit parameters.

The `Log / Logspaces / <logspace_name> / Maximum number of search results` setting limits the number of results the filter command can retrieve. If the result set exceeds this limit, an error message is received instead.

Command-specific response:

```
[
    {
        'processed_timestamp': <timestamp>,
        'timestamp': <timestamp>,
        'host': '<host name>',
        'program': '<program name>',
        'pid': <program pid>,
        'priority': <priority>,
        'facility': <facility>,
        'message': <message>,
        'tags': [],
        'dynamic_columns':
        {
            <column_name>: <value>,
            ...
        }
    },
    {
        'processed_timestamp': <timestamp>,
        'timestamp': <timestamp>,
        'host': '<host name>',
        ...
    },
    ...
]
```

Example using wget command:

To query the first ten log messages that contain the word 'starting' from the 'local' logspace without any restrictions on its reception time, use the following command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/filter/local?
from=0&to=9999999999&search_expression=starting&offset=0&limit=10'
```

Example for querying info from dynamic columns

To query logon messages collected by syslog-ng agent on windows (log messages contain 540 in .sdata.win@18372.4.event_id column) use a query similar to this:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/filter/center?
from=1389052800&to=1389099600&search_expression=nvpair:.sdata.win@18372.4.event_id=540
'
```

Note: the dynamic column is addressed with "nvpair" keyword as it is used in web UI.

/search/logspace/number_of_messages

Method: **GET**
Positional parameters:

logspace_name: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")

The command returns the number of messages that would be resulted by a query (filter command). The time interval and the search expression works the same way as in the filter command.

Command specific response:

```
<number_of_messages_corresponds_to_search_expression>
```

Example using wget command:

To query the number of log messages that contain the word 'starting' and program fields that contain 'syslog-ng' from the 'local' logspace without any restrictions on reception time, use the following command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/number_of_messages/local?
from=0&to=9999999999&search_expression=starting%20program:syslog-ng'
```

/search/logspace/generate_statistics

Method: **GET**
Positional parameters:

logspace_name: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")
column_name: Name of the column to generate statistics from; to generate statistics from a dynamic column, the prefix "dynamic:" must be used (required, regular expression: "(dynamic:.+)|priority|facility|host|program|pid|tag")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")
offset: Retrieve a subset of the result. The offset is the starting point of the subset. The value 0 means the first element in the result set (= no offsetting) (optional, default: "0", regular expression: "\d{1,19}")
limit: The maximum number of elements in the result set. If the whole result set has less elements than this value, then only those entries will be retrieved. More than 1000 records can be retrieved by executing several requests with consecutively increasing offset values (after an offset of 0, the next offset should be 1000, then 2000, and so on). (optional, default: "100", regular expression: "[0-9]{1,3}|1000")

Use the `generate_statistics` command to get statistics without querying, downloading and processing all the log messages. Statistics can be generated for one selected column from a given search query. The result is provided in descending order (it gives "top" type statistics). The number of elements in the statistics can be limited and an offset can be specified. Using offset together with the `number_of_statistics_entries` query, "least" type statistics can be generated as well.

Command specific response:

```
[
    [<value_in_requested_column_1>, <number_of_occurrences_of_value_1>],
    [<value_in_requested_column_2>, <number_of_occurrences_of_value_2>],
    [<value_in_requested_column_3>, <number_of_occurrences_of_value_3>],
    ...
]
```

Example using wget command:

To query the top 3 hosts generating messages that contain the words 'starting up" from the 'center' logspace without any restrictions on reception time, use the following com

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/generate_statistics/center/host?
from=0&to=9999999999&search_expression=starting%20up&offset=0&limit=3'
```

/search/logspace/number_of_statistics_entries

Method: **GET**
Positional parameters:

logspace_name: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")
column_name: Name of the column to generate statistics from; to generate statistics from a dynamic column, the prefix "dynamic:" must be used (required, regular expression: "(dynamic:.+)|priority|facility|host|program|pid|tag")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")

The `number_of_statistics_entries` command returns the total number of statistics entries for a given search query. This can be used to iterate through the list of different values if it is needed.

Command specific response:

```
<number_of_statistics_entries_for_query>
```

Example using wget command:

Retrieve the number of hosts that sent messages containing the words 'starting' in any time:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/number_of_statistics_entries/center/host?
from=0&to=9999999999&search_expression=starting'
```

/search/logspace/sum_of_statistics_values

Method: **GET**
Positional parameters:

logspace_name: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")
column_name: Name of the column to generate statistics from; to generate statistics from a dynamic column, the prefix "dynamic:" must be used (required, regular expression: "(dynamic:.+)|priority|facility|host|program|pid|tag")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")

The `sum_of_statistics_values` command returns the summarized value of statistical elements for a given column according to the search query. This can be used to calculate the relative frequency of a given value.

Command specific response:

```
<sum_of_statistics_values_for_query>
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/sum_of_statistics_values/center/host?
from=0&to=9999999999&search_expression=starting'
```

/search/logspace/list_dynamic_columns

Method: **GET**
Positional parameters:

> **logspace_name**: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")

SSB can store several types of information related to the log messages, accessible through dynamic columns. The `list_dynamic_columns` command returns the names of dynamic columns available for the given logspace.

Command specific response:

```
['<name_of_dynamic_column_1>', '<name_of_dynamic_column_2>', ...]
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/list_dynamic_columns/center'
```

/version

Method: **GET**

The `version` command returns the current version number of the API.

Command specific response:

```
<current_version>
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/version'
```

/search/logspace/number_of_messages_by_steps

Method: **GET**
Positional parameters:

> **logspace_name**: Name of the logspace in which to look for log messages (required, regular expression: "[a-zA-Z0-9\-_]+(\.?[a-zA-Z0-9\-_])*")

Get parameters:

from: Restrict the results to a given time interval; the from parameter specifies the start time for the time interval (required, regular expression: "\d{1,10}")
to: Restrict the results to a given time interval; the to parameter specifies the end time for the time interval; if to is in the future, the current time is used instead (required, regular expression: "\d{1,10}")
search_expression: Same search expression as used on SSB's web UI (optional, default: "", regular expression: ".*")
step: The amount of time between two queries (required, regular expression: "\d{1,10}")

The command returns an array of the number of messages that would be resulted by a query (filter command) in every specified second in the given time interval. The search expression works the same way as in the filter command.

Command specific response:

```
[
   <value_in_step_1>,
   <value_in_step_2>,
   <value_in_step_3>,
   ...
]
```

Example using wget command:

To query the number of log messages that contain the word 'starting' and program fields that contain 'syslog-ng' from the 'local' logspace, for every 1000 seconds in the given range use the following command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/number_of_messages_by_steps/local?
from=1389052800&to=1389099600&search_expression=starting%20program:syslog-
ng&step=1000'
```

/search/logspace/list_logspaces_details

Method: **GET**

Retrieves the list of searchable (indexed) logspaces, including the details of the logspace. Note that:

The list includes only those logspaces that the authenticated user can access.
The list includes only searchable (indexed) logspaces.
The list is in alphabetical order.

Command specific response:

```
[
    {"name" : "first_logspace_name", "is_alertable" : true},
    {"name" : "second_logspace_name", "is_alertable" : true},
    {"name" : "third_logspace_name", "is_alertable" : false},
    ...
]
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/list_logspaces_details'
```

/search/logspace/validate

Method: **GET**
Get parameters:

search_expression: Same search expression as used on SSB's web UI (required, regular expression: ".*")

Checks if the given search expression is valid.

Command specific response:

```
{"is_valid" : true}
```

Example using wget command:

```
wget -q -O - --header "Cookie: AUTHENTICATION_TOKEN=0123456789abcdef"
'https://192.168.10.48/api/4/search/logspace/validate?search_expression=starting'
```

Error codes

300
Description: Unknown command. Possible reasons: spelling mistake or wrong syntax.
HTTP error code: 404

350
Description: Wrong parameter(s). Possible reasons: missing, too many or wrong parameter(s).
HTTP error code: 404

400
Description: Authentication error. Possible reasons: wrong username or password.
HTTP error code: 403

450
Description: This user is not authorized to query the logspace.
HTTP error code: 403

451
Description: The config lock is currently held by another user or by the same user from another host
HTTP error code: 403

452
Description: This user is not authorized to import configuration bundle.
HTTP error code: 403

510
Description: Invalid argument.
HTTP error code: 500

520
Description: Logic error.
HTTP error code: 500

521
Description: Error with the validation of the uploaded config bundle.
HTTP error code: 500

522

Description: Error during the apply of the previously uploaded configuration bundle.
HTTP error code: 500

600

Description: Invalid enumeration.
HTTP error code: 500

700

Description: Validation exception.
HTTP error code: 500

800

Description: Unknown exception.
HTTP error code: 500